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Research Article

Analysis of contamination status and quantitative risk assessment of different sequence types of *Listeria monocytogenes* in ready-to-eat meat products

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Abstract: *Listeria monocytogenes* is a significant foodborne pathogen posing substantial risks to specific populations. This study investigated the contamination status and sequence types (ST) distribution of *L. monocytogenes* in ready-to-eat cooked meat products in the Shanghai region, and conducted quantitative risk assessments for major ST based on their growth parameters and dose-response parameters. A total of 675 ready-to-eat cooked meat product samples were collected, with 33 positive samples detected. ST8, ST5, ST87, ST9, and ST121 exhibited high detection rates, representing the distribution of moderate-, low-, and high-virulence strains. Risk assessment results showed that consumption of cooked meat products contaminated with moderate-virulence (ST5/ST8) strains caused the highest number of illnesses. The number of cases per million population increased with age, with more female patients aged 15–44 years old than male patients, and slightly more male patients than female patients over 44 years old. Pregnant women also faced elevated risks. Sensitivity analysis identified the initial pollution level, transportation temperature, and household storage temperature as the most significant factors affecting the final disease risk. This study provides a scientific basis for precise risk assessment and control of *L. monocytogenes*.

Keywords: *Listeria monocytogenes*; different sequence types; quantitative risk assessment; ready-to-eat foods; pregnant women

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1 Introduction

Listeria monocytogenes is a typical foodborne pathogen frequently associated with global public health incidents. It is the only pathogen capable of causing listeriosis in humans, which primarily manifests as meningitis, septicemia, miscarriage, and monocytosis. It poses a severe threat, particularly to pregnant women, the elderly, children, neonates, and immunocompromised individuals[1]. Compared with other foodborne pathogenic bacteria, the contamination rate of *L. monocytogenes* is relatively low, but the fatality rate and hospitalization rate it causes are relatively high. Among the 759 reported cases of *Listeria* infection in 22 provinces, municipalities, and autonomous regions in China from 2008 to 2017, 514 cases (68%) were related to pregnancy (including 116 pregnant women and 398 newborns), and the mortality rate in the newborn group was 24.3%[2]. According to the report of the Centers for Disease Control and Prevention of the United States, from 2004 to 2024, the United States reported about 1 600 cases of listeriosis each year, of which about 260 cases resulted in death. Sporadic and outbreak cases caused by *L. monocytogenes* are partly due to ready-to-eat cooked meat products, such as pre-packaged meat products and sausages[3]. The risk of consumers being infected with *L. monocytogenes* through ready-to-eat meat products is further increased due to the lack of a cooking step to inactivate bacteria before consumption and the pathogen's ability to grow under extreme conditions such as low temperature, low pH, and high salt concentration. The most severe *Listeria* outbreak in history, which occurred in 2017, was traced back to the consumption of polony, resulting in 937 reported cases[4].

L. monocytogenes can be divided into various subtypes, such as serotypes, lineages, multilocus sequence types (MLST), and clonal complexes (CC)[5]. There are significant differences in the distribution of ST of foodborne *L. monocytogenes* strains in different countries or regions. ST121 and ST9 are the two most common ST in Switzerland[6]. In Japan, CC6 (22%) strains are the most common among patients with listeriosis, followed by CC1 and CC2 strains[7]. In the United States and European countries, ST1, ST2, ST3, ST4, ST6, and ST8 strains are closely associated with clinical cases[7–10]. Our previous meta-analysis of the molecular typing of foodborne and clinical strains in the Chinese mainland from 2010 to 2019 found that among the *L. monocytogenes* strains in food, ST9 (39.77%), ST87 (12.03%), and ST8 (9.02%) accounted for a relatively high proportion. However, among the clinically isolated *L. monocytogenes* strains, ST87 (15.38%), ST8 (13.46%), and ST5 (11.54%) accounted for a higher proportion[8]. Among them, ST87 has become the most prevalent clinical source type in China, and it is rarely seen in other regions[8].

In the risk assessment process, strains variability should be considered to the greatest extent possible. The heterogeneity of strains in growth and inactivation characteristics can lead to differences in exposure levels across various exposure assessment stages, which in turn affects the final infection dose. Moreover, the virulence differences between different strains or ST can influence the dose-response relationship in hazard characterization, potentially leading to an overestimation or underestimation of the true risk of *L. monocytogenes* causing disease in the population. Currently, regarding strains variability, Pouillot et al.[11] analyzed the dose-response parameters for different CC types categorized by high, medium, and low virulence levels based on the relationship between strains in ready-to-eat foods and the prevalence of related illnesses, and established the dose-response relationships of different virulent strains. Based on ST or CC types, strains variability of *L. monocytogenes* in biofilm formation ability and virulence can be effectively distinguished[5,12]. Moreover, in practical risk assessment, once isolates are obtained, the ST information of the sample can be quickly determined. Therefore, if the probability of illness caused by *L. monocytogenes* can be derived based on ST, it would facilitate the rapid implementation of preventive and control measures. However, currently, there are no risk assessment studies specifically targeting different ST.

Therefore, in this study, we first investigated the contamination status and ST distribution of *L. monocytogenes* in ready-to-eat cooked meat products from the Shanghai region of China. Secondly, based on the growth parameters of different ST and the dose-response parameters of various virulence levels, a quantitative risk assessment of the major ST of *L. monocytogenes* in ready-to-eat cooked meat products was conducted. This assessment provides the disease probability of *L. monocytogenes* at different ST levels for different populations, offering a scientific basis for the precise risk assessment and prevention of *L. monocytogenes* in the future.

2 Materials and methods

2.1 Source of samples

A total of 675 pieces of cooked meat products were collected from 12 districts in Shanghai, China (2023–2024). The sampling areas covered the central urban area, the urban-rural junction area, and the far suburban area of the city, and the sampling locations included catering stores, retail stores, supermarkets, agricultural trade markets, and online stores. After the samples were collected, qualitative and quantitative tests were carried out on the samples plate counting according to *Microbiological examination of food: examination of Listeria monocytogenes* (GB 4789.30–2016).

2.2 Initial contamination level of *L. monocytogenes* in ready-to-eat cooked meat products in Shanghai, China

Among the 675 groups of cooked meat product samples collected, there were 33 positive samples and 642 negative samples. The initial positive contamination rate (P_p) is expressed in the form of a β distribution ($n_p + 1$, $n_n + 1$), where n_p is the number of positive samples and n_n is the number of negative samples[13–14]. That is, the P_p of *L. monocytogenes* in cooked meat products is β distribution (34, 676), and the negative contamination rate (P_n) can be expressed as 1 – P_p. The contamination concentration (L_p) of *L. monocytogenes* in positive samples is described by the cumulative distribution density function. The Jarvis equation (equation (1)) is used to calculate the median of negative contamination level (lg (CFU/g)) in negative samples, and the formula is as follows:

$$\text{Javis } M \text{ value} = \lg\left(-\frac{\ln P_n}{25}\right) \tag{1}$$

The concentration of *L. monocytogenes* in negative samples (L_n) is described by the cumulative distribution density function. The initial pollution distribution (N_0) of *L. monocytogenes* in cooked meat products is described by the discrete function: discrete (L_p:L_n, P_p:P_n).

2.3 Consumer handling parameters

This study assesses the risk of *L. monocytogenes* from the sales stage to the consumption stage, and there is no accurate data on the temperature curve during the transportation process from retail to households. Possas et al.[15] assumed that the transportation from retail to home can last at least 15 min and at most 2 h, so a uniform distribution (Uniform (0.24, 2)) was used. The temperature conditions during the transportation process remain largely undocumented, and it is generally observed that consumers do not utilize refrigeration for food during transit. At this stage, the temperature will rise, depending on many uncertain and variable factors. Nauta et al.[16] reported that while it is improbable for the temperature to increase from refrigerated levels to a maximum of 25 °C within 2 h, during hot weather conditions, rapid temperature escalation may occur; therefore, Pert (4, 10, 25) was used. According to the habits of residents, cooked meat products are stored in the refrigerator or at room temperature by residents and will be consumed within 24 h. Therefore, the Pert (4, 10, 25) and Pert (0.48, 1.92, 24) distributions are used to describe the household storage temperature and storage time respectively.

2.4 Construction of growth prediction model of *L. monocytogenes*

The maximum specific growth rate (μ_{max}) of different ST (ST8, ST9, ST121, ST5, and ST87) strains was measured. The working stock solution of *L. monocytogenes* was stored at 4 °C in tryptic soy agar containing 0.6% (*m*/*m*) yeast extract (TSA-YE; Hopebio, Qingdao, China) and updated monthly. To activate each strain before the experiment, a single colony on a TSA-YE plate was inoculated into 10 mL of tryptic soy broth with yeast extract (TSB-YE; Hopebio, Qingdao, China) and incubated at 37 °C for 18–24 h before each experiment. The prepared bacterial suspension was diluted to 10^3, 10^4, 10^5, 10^6, and 10^7 CFU/mL using TSB-YE. The 200 μL of these bacterial solutions were added to each well of a 100-well honeycomb plate, while 200 μL of TSB-YE was used as a blank control group. The plate was then placed in the sample chamber of a Bioscreen C (Oy Growth Curves Ab Ltd., Helsinki, Finland) and statically incubated at 37 °C for 24 h, with $OD_{600\text{ nm}}$ measured every 30 min. Before each measurement, the plate was shaken at medium intensity for 20 s. These procedures were repeated 3 times with 2 replicates. The μ_{max} was calculated from the obtained turbidity growth curves using the time to detection method[17]. When the $OD_{600\text{ nm}}$ of the bacterial suspension reached 0.17 (with an initial $OD_{600\text{ nm}}$ of 0.10), the concentration of the bacterial solution was considered to be approximately 10^7 CFU/mL, and this time point was recorded as t_d (h). The μ_{max} was calculated using the following formula (2):

$$\mu_{\max} = -\frac{1}{k} \tag{2}$$

Where, k represents the slope of the linear regression curve between the natural logarithms of different initial concentrations (10^3, 10^4, 10^5, 10^6, and 10^7 CFU/mL) and t_d at these different initial concentrations.

The Baranyi and Roberts model (equations (3) and (4)) is selected as the primary growth model[18]:

$$A(t) = t + \frac{1}{\mu_{\max}} \ln\left[\exp(-\mu_{\max} t) + \exp(-\mu_{\max}\lambda) - \exp(-\mu_{\max} t - \mu_{\max}\lambda)\right] \tag{3}$$

$$N(t) = N_0 + \mu_{\max} A(t) - \ln\left[1 + \frac{\exp(\mu_{\max} A(t)) - 1}{\exp(N_{\max} - N_0)}\right] \tag{4}$$

Where, $N(t)$ represents the final contamination level of *L. monocytogenes* within time t (ln (CFU/g)); t represents the growth time of *L. monocytogenes* (h); N_0 represents the initial pollution distribution of *L. monocytogenes* in cooked meat products (discrete (L_p:L_n, P_p:P_n)) (ln (CFU/g)); $N_{\max}$ represents the maximum growth value of the microbial population (ln (CFU/g)); $\mu_{\max}$ represents the maximum specific growth rate of *L. monocytogenes* (h^{-1}); λ represents the lag time (h).

In this study, the γ model was adopted as the secondary growth model for *L. monocytogenes*[19]. By estimating the actual temperature (equation (5)), water activity (a_w) (equation (6)) and pH value (equation (7)), the $\mu_{\max}$ under limited conditions was calculated, and the formula is shown in equation (8):

$$\gamma(T) = \left(\frac{T - T_{\min}}{T_{\text{opt}} - T_{\min}}\right)^2 \tag{5}$$

$$\gamma(a_w) = \frac{a_w - a_{w\min}}{1 - a_{w\min}} \tag{6}$$

$$\gamma(\text{pH}) = \frac{(\text{pH} - \text{pH}_{\min}) \times (\text{pH}_{\max} - \text{pH})}{(\text{pH}_{\text{opt}} - \text{pH}_{\min}) \times (\text{pH}_{\max} - \text{pH}_{\text{opt}})} \tag{7}$$

$$\mu_{\max} = \mu_{\max\,\text{opt}} \times \gamma(T) \times \gamma(a_w) \times \gamma(\text{pH}) \tag{8}$$

2.5 Construction of dosage-response relationship model

The exponential model assumes that each pathogen has the same infection probability, and within the low-dose range, the infection probability shows a linear relationship with the dose. It is suitable for describing low-dose exposure scenarios where pathogens have strong virulence and hosts exhibit high sensitivity to the pathogens[20]. Compared with the β-Poisson model, the exponential model is more straightforward in mathematical form and relatively easier for parameter estimation. In this study, through the fitting of actual data and statistical tests, the exponential model can better reflect the relationship between different ST of *L. monocytogenes* and infection risk under low-dose exposure. Therefore, this study adopted an exponential dosage-response model, which is as follows:

$$\text{Prob} = 1 - \exp(-r \times d) \tag{9}$$

Where, Prob represents the probability of invasive listeriosis; r represents the pathogenic probability of a single *L. monocytogenes* cell, and lg r follows a normal distribution Normal (μ, σ), μ is the mean and σ is the standard deviation; d is the exact number of ingested bacteria (CFU). In this study, the parameters σ and μ obtained by Pouillot et al.[11] based on different populations and different virulence levels were used.

ST is based on MLST technology, which classifies strains by analyzing nucleotide sequence differences in multiple housekeeping genes[5]. This approach provides more detailed genetic information and accurately reflects the evolutionary relationships and virulence differences among strains[21]. In contrast, serotyping is based solely on differences in surface antigens, resulting in lower typing resolution. Therefore, ST is more suitable as a basis for virulence classification. In this study, the virulence category classification defined by Fritsch et al.[22] was used. These categories are named low virulence, moderate virulence, and high virulence. Fritsch et al.[22] adopted the clinical frequency data compiled by Maury et al.[23] and identified 6 633 strains of *L. monocytogenes* isolated from clinical samples and foods. The clinical frequency was calculated by dividing the number of clinical isolates of a specific CC by the total number of clinical and food isolates of that CC. Therefore, in this study, ST9 and ST121 were classified as low virulence, ST8 and ST5 were categorized as medium virulence, and ST87 was classified as high virulence.

Exposure assessment was carried out by considering 10 age-gender subgroups (age categories: 15–24, 25–44, 45–64, 65–74, and ≥ 75 years old, and gender categories: male and female) (Table 1), as well as pregnant women as population groupings. For the single population of pregnant women, μ is −11.70[24], and the value of σ in this project is calculated according to the virulence grade.

Table 1 The μ and σ of normal distribution parameters for lg r values across 10 population subgroups and three virulence levels.

Population		Low virulence (σ = 1.247)	Moderate virulence (σ = 0.836)	High virulence (σ = 0.756)
15–24 years	Female	−14.166	−12.296	−11.671
	Male	−14.124	−12.256	−11.625
25–44 years	Female	−14.516	−12.582	−12.046
	Male	−14.633	−12.690	−12.165
45–64 years	Female	−14.002	−12.123	−11.521
	Male	−14.668	−12.730	−12.200
65–74 years	Female	−13.708	−11.815	−11.239
	Male	−14.444	−12.522	−11.973
> 75 years	Female	−13.755	−11.890	−11.272
	Male	−13.753	−11.869	−11.274

2.6 Scoring criteria for the grading of pathogenic potential

The scoring criteria are as follows: 1) culture conditions: for a strain of a specific virulence, if either the $\mu_{\max}$ or λ index is the highest, 1 point is awarded; if the second-highest, 0.5 points are awarded; and if the lowest, no points are awarded. 2) Genes: if the *Listeria* pathogenicity-3 (LIPI-3) virulence island is present, 1 point is awarded. If the *Listeria* pathogenicity islands-4 (LIPI-4) virulence island is present, 1 point is awarded. If no premature stop codon (PMSC) is present in *inlA*, 1 point is awarded. 3) Virulence: if its adhesion rate, invasion rate, or barrier-crossing ability to cells is the highest, 1 point is awarded; if the second-highest, 0.5 points are awarded; and if the lowest, no points are awarded. 4) Risk assessment: for the annual number of listeriosis cases per million population in different subgroups, 3 points are awarded for the highest incidence, 2 points for the second-highest, and 1 point for the lowest.

2.7 Data processing

Microsoft Excel 2016 software was applied for data analysis, and the @Risk 5.7 software (Palisade Corp., Ithaca, New York) was used to conduct a simulation analysis of the parameters in the model. All preset parameters of the probability distribution forms were obtained by using the Monte Carlo simulation sampling method and through 100 000 iterations. The parameters of *L. monocytogenes* in each link of cooked meat products and the evaluation model are shown in Table 2. The model uses the probability distribution function to represent the uncertain values of variables. In the exposure stage of the risk assessment, the average value of the simulated values was substituted into the output value. The generated distribution is expressed in terms of the mean value, standard deviation, median, and the 5th and 95th percentiles. At the same time, the Spearman correlation coefficient was used to conduct a sensitivity analysis between the low variables

Table 2 Parameters for the exposure assessment of *L. monocytogenes* at different stages in cooked meat products.

Parameter	Unit	Description	Calculations/distribution			Reference
			Low virulence	Moderate virulence	High virulence	
$\mu_{\text{max opt}}$	ln (CFU/(h·g))	Maximum specific growth rate	1.037	1.093	1.090	
T_{min}	°C	Minimum growth temperature		−1.72		[25]
T_{opt}	°C	Optimal growth temperature		37		[25]
pH_{min}		The lowest pH value		4.71		[25]
pH_{max}		The maximum pH value		9.61		[25]
pH_{opt}		Optimal pH value		7.1		[25]
pH_{meat}		The pH value of cooked meat products		Uniform (5.4, 6.95)		[26]
$a_{\text{w min}}$		Minimum a_w		0.913		[25]
$a_{\text{w meat}}$		a_w of cooked meat products		Uniform (0.944, 0.985)		[26]
a_0		Initial value of physiological state		Pert (0, 0.02, 1)		[27]
Javis *M* value	lg (CFU/g)	The median of negative contamination level in negative samples		Equation (1)		
N_{max}	lg (CFU/g)	Maximum growth concentration		Pert (3.6, 8.2, 8.5)		[28]
P_p	%	The initial positive contamination rate	Beta (7, 676)	Beta (20, 676)	Beta (5, 676)	
P_n	%	The negative contamination rate		$1 - P_p$		
L_p	lg (CFU/g)	The positive contamination concentration	Cumul (1, 2.64, {1, 2.64}, {0.67, 1})	Cumul (1, 2.76, {1, 2.34, 2.57, 2.6, 2.66, 2.69, 2.76}, {0.22, 0.33, 0.44, 0.56, 0.67, 0.78, 1})	Cumul (1, 2.83, {1, 2.52, 2.83}, {0.33, 0.67, 1})	
L_n	lg (CFU/g)	The negative contamination concentration	Cumul (−7.24, 0.48, {−7.24, −3.38, 0.48}, {0.01, 0.5, 0.99})	Cumul (−6.32, 0.48, {−6.32, −2.92, 0.48}, {0.01, 0.5, 0.99})	Cumul (−7.54, 0.48, {−7.54, −3.53, 0.48}, {0.01, 0.5, 0.99})	
N_0		The initial pollution distribution		Discrete ($L_{\text{p China}}$:$L_{\text{n China}}$, $P_{\text{p China}}$:$P_{\text{n China}}$)		[29]
t_1	h	Retail stage time		Pert (0, 24, 96)		[27]
T_1	°C	Retail storage temperature		Normal (5.44, 2.32)		[30]
μ_{max1}	ln (CFU/(g·h))	Growth rate in the retail stage		Equations (5)–(8)		[30]
λ_1	h	Lag period		$(-\ln a_0)/\mu_{\text{max1}}$		
$A(t)_1$		Model parameter		Equation (3)		
N_1	ln (CFU/g)	The pollution level at the end of retail		Equation (4)		
t_2	h	From retail to home delivery time		Uniform (0.24, 2)		[15]
T_2	°C	From retail to household transportation temperature		Pert (4, 10, 25)		[16]
μ_{max2}	ln (CFU/(g·h))	Bacterial growth rate during transportation		Equations (5)–(8)		[30]
λ_2	h	Lag period		$(-\ln a_0)/\mu_{\text{max2}}$		
$A(t)_2$		Model parameter		Equation (3)		
N_2	ln (CFU/g)	Pollution level at the transport destination		Equation (4)		
t_3	h	Storage time in the consumption stage		Pert (0.48, 1.92, 24)		Expert opinion
T_3	°C	Storage temperature during the consumption stage		Pert (4, 10, 25)		[27]
μ_{max3}	ln (CFU/(g·h))	The growth rate of bacteria during the consumption stage		Equations (5)–(8)		[30]
λ_3	h	Lag period		$(-\ln a_0)/\mu_{\text{max3}}$		
$A(t)_3$		Model parameter		Equation (3)		
N_3	ln (CFU/g)	Pollution level at the consumption end		Equation (4)		
C_{RTE}	g/(day·person)	Daily consumption amount of ready-to-eat cooked meat		Pert (0, 24.33, 463.01)		This study
$\text{Dose}_{\text{meat}}$	CFU	The dosage of the ingested bacteria		$C_{\text{RTE}} \times e^{N3}$		[31]
lg *r*				Normal (μ, σ)		[11]
Prob		The probability of invasive listeriosis		$1 - \exp(-r \times \text{Dose}_{\text{meat}})$		[31]
Prob_{100}		The number of patients per million population		$(1 - (1 - P)^{365}) \times 10^6$		[31]

and the results of the model exposure assessment, and appropriate management measures were proposed.

3 Results and discussion

3.1 Sampling data of cooked meat products

In Shanghai from 2023 to 2024, a total of 675 samples of cooked meat products were collected, including cooked livestock meat, cooked poultry meat, ham sausages, luncheon meat, dried meat floss, and cured meat. The detection rates of ST8, ST5, ST87, ST9, and ST121 were relatively high (Table 3). The detection rate of *L. monocytogenes* in cooked meat products was 4.89% (33/675), and the quantitative detection results of 18 groups of data were obtained (Table 4). The proportion of moderately virulent strains (ST5/ST8) was 66.67%, and the average concentration of moderately virulent strains in positive samples (33 samples) was 2.17 (lg (CFU/g)), which was significantly higher than that of highly virulent strains (1.81 (lg (CFU/g))) (Table 5).

Table 3 The detection numbers and detection rates of different ST of *L. monocytogenes* in cooked meat products.

ST	Detection number	Detection rate (%)
ST1	2	6.06
ST10	1	3.03
ST121	4	12.12
ST3	1	3.03
ST5	8	24.24
ST602	1	3.03
ST8	11	33.33
ST87	4	12.12
ST9	2	6.06

Table 4 Summary of quantitative detection results of *L. monocytogenes* of different ST in cooked meat products.

Strain No.	Number of colonies (CFU)	Colony concentration (CFU/(g·mL))	MLST
L1	440	2.64	ST9
L2	680	2.83	ST87
L3	330	2.52	ST87
L4	< 10	1.00	ST87
L5	400	2.60	ST8
L6	490	2.69	ST8
L7	< 10	1.00	ST8
L8	< 10	1.00	ST8
L9	370	2.57	ST5
L10	570	2.76	ST5
L11	220	2.34	ST5
L12	580	2.76	ST5
L13	460	2.66	ST5
L14	< 10	1.00	ST121
L15	< 10	1.00	ST121
L16	< 10	1.00	ST10
L17	520	2.72	Unconfirmed
L18	< 10	1.00	Unconfirmed

Table 5 Virulence distribution and detection rates of different ST of *L. monocytogenes* in cooked meat products.

Virulence category	Low virulence	Moderate virulence	High virulence
ST	ST9, ST121	ST8, ST5, ST3, ST10, ST602	ST87, ST1
Detection rate (%)	18.18	66.67	18.18

3.2 Consumption data of cooked meat products in Shanghai, China

A survey on the frequency of consumption of cooked meat products was conducted among people over 15 years old in Shanghai in 2023. A multi-stage stratified random sampling method was adopted to select the sample population in the central urban area, urban-rural fringe area, and outer suburban area of the city for the survey. Eventually, 2 900 valid samples were obtained. The survey results showed that the average daily consumption of cooked meat products per capita among Shanghai residents was 24.33 g, with a minimum consumption is 0 g/day, a maximum consumption is 463.01 g/day, and P_{95} is 89.19 g/day.

3.3 Exposure assessment of *L. monocytogenes* in cooked meat products

Exposure assessment is mainly divided into three modules: retail, transportation, and home consumption. By simulating the changes in the bacterial concentration of *L. monocytogenes* under specific links, the exposure concentration is obtained. The distribution of the initial contamination level of *L. monocytogenes* in cooked meat is shown in Table 6. The average contamination concentration of low-virulence (ST121/ST9) *L. monocytogenes* in cooked meat is −3.33 (lg (CFU/g)), with a 90% confidence interval (CI) of −6.92–0.24 (lg (CFU/g)); the average contamination concentration of moderately-virulent (ST5/ST8) *L. monocytogenes* is −2.77 (lg (CFU/g)), with a 90% CI of −6.03–0.40 (lg (CFU/g)); the average contamination concentration of highly-virulent (ST87) *L. monocytogenes* is −3.49 (lg (CFU/g)), with a 90% CI of −7.21–0.21 (lg (CFU/g)).

The fitting of the growth situation of *L. monocytogenes* in cooked meat products at the retail stage is shown in Table 6. After retail stage, the average contamination concentration of low-virulence (ST121/ST9) *L. monocytogenes* in cooked meat is −3.10 (lg (CFU/g)), with a 90% CI of −6.75–0.47 (lg (CFU/g)); the average contamination concentration of moderately-virulent (ST5/ST8) *L. monocytogenes* is −2.52 (lg (CFU/g)), with a 90% CI of −5.86–0.73 (lg (CFU/g)); the average contamination concentration of highly-virulent (ST87) *L. monocytogenes* is −3.24 (lg (CFU/g)), with a 90% CI of −7.03–0.46 (lg (CFU/g)).

L. monocytogenes in cooked meat products showed an increasing trend during the transportation process. After transportation, the average contamination concentration of low-virulence (ST121/ST9) *L. monocytogenes* in cooked meat is −3.09 (lg (CFU/g)), and the maximum concentration reaches 0.49 (lg (CFU/g)) (95% CI); the average contamination concentration of moderately-virulent (ST5/ST8) *L. monocytogenes* is −2.51 (lg (CFU/g)), and the maximum concentration reaches 0.75 (lg (CFU/g)) (95% CI); the average contamination concentration of highly-virulent (ST87) *L. monocytogenes* is −3.23 (lg (CFU/g)), and the maximum concentration reaches 0.47 (lg (CFU/g)) (95% CI).

During home storage, cooked meat is generally stored refrigerated in the refrigerator or room temperature. The concentration of

Table 6 The contamination levels (lg (CFU/g)) of *L. monocytogenes* at different stages in cooked meat products.

L. monocytogenes strains	Outputs	Minimum	Maximum	Mean	Standard deviation	5% CI	95% CI
Low virulence (ST121/ST9)	N_0	−7.24	2.63	−3.33	2.31	−6.92	0.24
	N_1 (*L. monocytogenes* level after retail)	−7.24	5.32	−3.10	2.34	−6.75	0.47
	N_2 (*L. monocytogenes* level after transportation)	−7.24	5.36	−3.09	2.34	−6.74	0.49
	N_3 (*L. monocytogenes* level after home storage)	−7.24	5.85	−2.99	2.36	−6.66	0.58
Moderate virulence (ST5/ST8)	N_0	−6.32	2.76	−2.77	2.15	−6.03	0.40
	N_1 (*L. monocytogenes* level after retail)	−6.32	6.23	−2.52	2.19	−5.86	0.73
	N_2 (*L. monocytogenes* level after transportation)	−6.32	6.28	−2.51	2.19	−5.84	0.75
	N_3 (*L. monocytogenes* level after home storage)	−6.32	6.67	−2.40	2.21	−5.76	0.98
High virulence (ST87)	N_0	−7.54	2.83	−3.49	2.40	−7.21	0.21
	N_1 (*L. monocytogenes* level after retail)	−7.54	7.77	−3.24	2.43	−7.03	0.46
	N_2 (*L. monocytogenes* level after transportation)	−7.54	7.77	−3.23	2.43	−7.02	0.47
	N_3 (*L. monocytogenes* level after home storage)	−7.54	7.79	−3.12	2.45	−6.94	0.57

L. monocytogenes in cooked meat increased after storage. After home storage, the average contamination concentration of low-virulence (ST121/ST9) *L. monocytogenes* in cooked meat is −2.99 (lg (CFU/g)), and the maximum concentration reaches 0.58 (lg (CFU/g)) (95% CI); the average contamination concentration of moderately-virulent (ST5/ST8) *L. monocytogenes* is the highest, which is −2.40 (lg (CFU/g)), and the maximum concentration reaches 0.98 (lg (CFU/g)) (95% CI); the average contamination concentration of highly-virulent (ST87) *L. monocytogenes* is the lowest compared with the other two virulent strains, which is −3.12 (lg (CFU/g)), and the maximum concentration reaches 0.57 (lg (CFU/g)) (95% CI). Therefore, at the end point of the exposure assessment, that is, after home storage, the contamination level of moderately-virulent (ST5/ST8) *L. monocytogenes* in cooked meat is the highest.

3.4 Risk estimation

By applying the dose-response model, Table 7 lists the annual number of listeriosis cases per million population in different subgroups due to the consumption of cooked meat products. For the low-virulence (ST121/ST9) strains, the average number of cases per million population is relatively low in most subgroups. In the moderate-virulence strain group, the number of cases is more prominent. Among females aged 25–44 and the population aged 45–64, the number of cases is higher than that of the low-virulence (ST121/ST9) and high-virulence (ST87) strains. Especially in the elderly population, for example, in the age group over 75 years old, the average number of cases per million population is 112.3 for females and 76.5 for males. The high-virulence strains (ST87) also continuously show a relatively high disease burden, with an average number of cases higher than that of other virulence strains in the younger population (such as females aged 15–24, with an average of 33.8 cases).

In addition, the gender difference in the number of cases per million population is also very obvious. Among the elderly population over 44 years old, when infected with moderate-virulence and high-virulence strains, the number of male cases is slightly more than that of female cases. This may be related to the higher proportion of bad living habits (such as heavy drinking and smoking) among men in this age group, as well as the gradual decline of immune system function with age, particularly the relatively faster decline rate in men. These factors may lead to a decrease in men's resistance to *L. monocytogenes*, thereby increasing the risk of infection[32]. While among the population aged 15–44, the number of female cases is higher than that of male cases. This may be because the age group of 15–44 includes women of childbearing age, and pregnant women are a susceptible population to *L. monocytogenes*. During pregnancy, changes in hormone levels lead to adjustments in immune system function, increasing pregnant women's susceptibility to *L. monocytogenes*. Additionally, the relatively large proportion of women of childbearing age in the population contributes to the elevated overall infection rate among females in this age group[33].

Table 7 Distribution of the annual number of listeriosis cases per million population in different subgroups due to the consumption of cooked meat products.

Age (year)/type	Gender	Low virulence (ST121/ST9)		Moderate virulence (ST5/ST8)		High virulence (ST87)	
		Mean	95% CI	Mean	95% CI	Mean	95% CI
15–24	Female	0.1	0.0	14.8	0.6	33.8	0.7
	Male	0.1	0.0	10.1	0.1	16.3	0.1
25–44	Female	1.2	0.0	43.5	1.2	36.8	1.3
	Male	0.0	0.0	8.8	0.2	16.5	0.2
45–64	Female	0.1	0.0	36.5	1.0	27.3	1.3
	Male	0.1	0.0	49.7	1.1	28.7	1.2
65–74	Female	0.2	0.0	60.5	2.1	43.5	2.8
	Male	0.3	0.0	72.1	2.8	47.3	3.8
> 75	Female	0.6	0.0	112.3	3.0	58.6	4.3
	Male	0.4	0.0	76.5	2.7	59.8	4.1
Pregnant women	Female	11.3	1.2	46.7	1.6	24.4	0.5

For the single group of pregnant women, the number of cases per million population infected with low-virulence (ST121/ST9), moderate-virulence (ST5/ST8), and high-virulence (ST87) strains is 11.3, 46.7, and 24.4, respectively (Table 7).

3.5 Sensitivity analysis

A sensitivity analysis was conducted with the annual number of listeriosis cases per million population. The study found that for the low-virulence (ST121/ST9) *L. monocytogenes* strains, the influencing factors of the number of patients in different age groups and genders are different. Overall, however, the temperature during the transportation stage and the initial contamination concentration has a greater impact. For the moderately-virulent (ST5/ST8) and highly-virulent (ST87) strains, the influencing factors of the number of patients in different age groups and genders are similar. The factor with the greatest impact is the initial contamination concentration, followed by the temperature during the transportation stage and the temperature during the home storage stage.

The above results indicate that the initial contamination level of *L. monocytogenes* in cooked meat products, as well as the transportation and home storage temperatures, are the key factors determining the final number of listeriosis cases. Therefore, environmental hygiene supervision should be strengthened, and the operation process should be standardized in the processing, transportation, and storage of cooked meat products to effectively control the initial contamination level. In addition, attention should be paid to the storage temperature and time during the transportation and home storage processes, and the cooked meat products should be consumed as soon as possible. The home storage time should be minimized to the greatest extent to reduce the risk of getting sick.

3.6 Uncertainty analysis

This assessment is conducted for *L. monocytogenes* in cooked meat products, and the main uncertainties in the assessment were analyzed. The dose-response relationship is uncertain. The risk assessment of *L. monocytogenes* refers to the parameters of high, medium and low virulence in the exponential dose-response model published by the European Food Safety Authority. Currently, there is no dose-response model for different populations established based on the disease prevalence data in China, which increases the uncertainty of risk prediction. In addition, there is also uncertainty in the matrix of ready-to-eat cooked food products. In this assessment work, there are many types of cooked meat products. Although parameters such as pH value and a_w have been set, other components in cooked meat products, such as nitrate concentration, phenol concentration, lactic acid concentration, preservative concentration, and bacteriostatic agent concentration, can also affect the growth of *L. monocytogenes*. Nitrate can inhibit the growth of *Listeria* by decomposing to produce nitric oxide, which interferes with the energy metabolism and DNA synthesis of *L. monocytogenes*[34]. Preservatives such as nisin can damage the cell membrane of *L. monocytogenes*, leading to the leakage of intracellular substances and thus inhibiting bacterial reproduction[35]. Potassium sorbate inhibits the growth of *L. monocytogenes* by reducing the intracellular pH value, inhibiting the enzyme activity and metabolic processes of *L. monocytogenes*[36]. Therefore, the actual survival amount in the processing link should be lower. There is also uncertainty in the household consumption link: this assessment does not take into account the cross-contamination in the household consumption link.

3.7 Grading of pathogenic potential

Based on the literature data and the morbidity probabilities of different virulence obtained in this study, a simple virulence grading score was further conducted for different ST of *L. monocytogenes*. The pathogenic potential of *L. monocytogenes* with different ST is scored from four aspects, and each aspect has the same weight. The pathogenic potential of *L. monocytogenes* with different ST was ranked according to its growth characteristics, virulence, the distribution of pathogenic potential genes, and the results of the risk assessment, as shown in Table 8. When a certain ST strains shows strong pathogenic potential in a certain aspect, a specified score can

Table 8 Scoring results of the pathogenic potential of *L. monocytogenes* with different ST.

Parameter	Scoring perspective	Score	Low virulence (ST121/ST9)	Moderate virulence (ST5/ST8)	High virulence (ST87)	Reference
Growth (1)	TSB-YE	1	0	1	0.5	This study
Gene (2)	LIPI	1	0	0	1	[12]
	inlA PMSC	1	0	1	1	[12]
Virulence (3)	The ability to translocate across the Caco-2 cells barrier	1	0.5	0	1	[12]
	Adhesion ability of JEG-3 cells	1	0	1	0.5	[12]
	Invasion ability of JEG-3 cells	1	0	1	0.5	[12]
Risk assessment (4)	15–24 years female	3	1	2	3	This study
	15–24 years male	3	1	2	3	This study
	25–44 years female	3	1	3	2	This study
	25–44 years male	3	1	2	3	This study
	45–64 years female	3	1	3	2	This study
	45–64 years male	3	1	3	2	This study
	65–74 years female	3	1	3	2	This study
	65–74 years male	3	1	3	2	This study
	> 75 years female	3	1	3	2	This study
	> 75 years male	3	1	3	2	This study
	Total	36	10.5	31	27.5	

be added. Thus, the risks of *L. monocytogenes* are preliminarily and exploratively classified from high to low according to the scores.

The moderate-virulence (ST5/ST8) strains of *L. monocytogenes* exhibit the strongest growth characteristics in TSB-YE medium, and has a complete *inlA* gene coding internalin A (InlA). InlA is described as the main invasive protein responsible for the uptake of *L. monocytogenes* by normal non-phagocytic cells, mediating bacterial entry into cells[37]. The mutation of *inlA* (such as the occurrence of a PMSC) is associated with a decrease in strain invasiveness[12]. Different ST of *L. monocytogenes* exhibit distinct *inlA* expression. Song et al.[12] reported that ST121 and ST9 carry PMSCs in *inlA*, while most other ST (e.g., ST8, ST5, ST87, ST1930) possess full-length *inlA* genes. Moreover, ST5 strains have a strong show strong adhesion and invasion capabilities to placental JEG-3 cells[12]. Risk assessment reports on *L. monocytogenes* in cooked meat products indicate that moderate-virulence strains (ST5/ST8) display higher case numbers per million population across most age groups. Therefore, the moderate-virulence strains (ST5/ST8) strains have the highest pathogenic potential score of 31 points. The high-virulence (ST87) strains of *L. monocytogenes* did not exhibit the highest growth characteristics in TSB-YE medium, but they contains the *Listeria* LIPI-4 virulence island and showed a strong ability to cross the intestinal cell barrier[12]. The presence of LIPI-4 has been proven to be closely associated with increased virulence and placental infection. Studies have shown that gene knockout within LIPI-4 can downregulate the transcription of several *L. monocytogenes* virulence genes, thereby affecting the virulence of the strain[38]. Most *L. monocytogenes* strains do not contain LIPI-4, which is potentially carried by CC4 and CC87 strains[39]. Risk assessment reports on *L. monocytogenes* in cooked meat products indicate that the high-virulence (ST87) strains have higher average case numbers in younger populations compared to other virulence strains. Therefore, the pathogenic potential of high-virulence (ST87) strains is second only to that of moderate-virulence (ST5/ST8), with a pathogenic potential score of 27.5 points. The low-virulence (ST121/ST9) strains carry PMSC in *inlA*, exhibit the lowest ability to infect cells and the lowest number of cases per million population in humans. Therefore, they have the lowest pathogenic potential score of 10.5 points (Table 8).

4 Conclusion

This study systematically investigated the contamination status of different ST of *L. monocytogenes* in ready-to-eat cooked meat products in the Shanghai region and conducted quantitative risk assessments. It was confirmed that ST8, ST5, ST87, ST9, and ST121 are the dominant ST, with moderate-virulence, low-virulence, and high-virulence strains all present in ready-to-eat cooked meat products at certain proportions. Through risk assessment, this study revealed differences in pathogenic risks of ST strains with different virulence levels across various populations, as well as the impact of age and gender on infection risks. However, this study only analyzed some common ST. Future research should expand the coverage of ST by incorporating more rare or novel ST strains to comprehensively understand the genetic diversity and risk characteristics of *L. monocytogenes*, thereby making risk assessment results more universally applicable. Currently, the parameters of the dose-response model used in risk assessments mostly reference foreign data, which may differ from actual conditions in China. In the future, localized dose-response models should be developed based on domestic population disease prevalence data to reduce uncertainties in risk predictions and improve the accuracy of risk assessments. This assessment did not fully consider cross-contamination at various stages from food production to consumption, nor did it adequately investigate the impact of food matrix components. Subsequent research needs to deeply analyze the microbial dynamics throughout the entire food production, processing, transportation, storage, and consumption chain, comprehensively considering the effects of cross-contamination and food components on the growth and survival of *Listeria*, and refining risk assessment models to provide more comprehensive support for food safety assurance.

Conflict of interest

The authors declare that they have no conflict of interest.

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